

ASHWATH GURU

Data & AI Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 7+ years of experience building and leading large-scale batch and streaming data pipelines in Scala, Python, and SQL using Apache Spark, Databricks, and Delta Lake. Proven track record designing lakehouse architectures (Apache Iceberg, Delta Lake) and distributed MPP query engines (Trino) that power engineering, analytics, and data science teams at scale. Experienced across the full pipeline SDLC orchestration (Argo Workflows), observability, data quality, lineage, logging, and alerting to protect strict uptime SLAs and deliver a single, reliable source of truth. Comfortable operating as a subject-matter expert across domain data areas, collaborating with product managers, architects, and analytics stakeholders, and mentoring engineers within an Agile/Scrum environment.

CORE TECHNICAL SKILLS

Languages: Python, Scala, SQL, PL/SQL
Big Data & Lakehouse: Apache Spark, PySpark, Databricks, Delta Lake, Apache Iceberg, Trino (MPP query engine), Hive
Pipeline Orchestration: Argo Workflows, Argo CD (Airflow-equivalent paradigm), Kubernetes, Docker, Jenkins, CI/CD
Cloud Platforms: AWS (S3, Glue, Lambda), Google Cloud Platform (GCP), Azure Data Factory
Data Quality & Observability: Instana, pipeline logging & alerting, data lineage, SLA monitoring
Streaming & Messaging: Apache Kafka, OpenSearch
Databases & Warehousing: PostgreSQL, NoSQL, Oracle, Data Modeling, Data Warehousing
AI / GenAI: LLMs, LangChain, OpenAI API, Hugging Face, FAISS, RAG, Agentic APIs
Practices: REST API Design, ETL/ELT Pipeline Design, Agile/Scrum, Technical Documentation & Data Governance

PROFESSIONAL EXPERIENCE

Data & AI Engineer | IBM – ISL, OMS-SIP May 2025 – Present

- Leading the build of “Data Insights,” a new data platform from the ground up on top of a multi-petabyte Apache Iceberg data lake, establishing foundational pipelines and datasets consumed by both business users and AI agents through a unified, query-driven API layer.
- Designed a query abstraction service that translates user- and agent-issued query requests into optimized queries against the distributed MPP engine Trino over Iceberg tables, returning real-time insights on metrics such as stagnant SKUs and node performance under strict latency SLAs.
- Built a parameterized, self-serve batch reporting pipeline that lets users and AI agents request custom reports on demand, with results delivered in CSV format — maintaining data accuracy and reliability for downstream analytics and operations teams.
- Developed distributed batch data processing jobs in Python and PySpark to transform and prepare large-scale datasets for low-latency query access, working closely with product and analytics stakeholders to define data contracts.
- Own end-to-end SDLC for pipeline changes: orchestrated and deployed jobs using Argo Workflows and Argo CD on Kubernetes (DAG-based orchestration analogous to Airflow), containerized with Docker, with a fully automated CI/CD workflow.
- Integrated Apache Kafka for event-driven streaming ingestion and OpenSearch for indexing and search analytics, extending the platform beyond batch into near-real-time processing.
- Built out observability, logging, and alerting using Instana to protect pipeline uptime SLAs, and used PostgreSQL for service metadata, configuration, and lineage tracking; maintain documentation to support data quality and governance requirements.
- Mentor engineers on the team on data platform architecture, Spark job design, and debugging production pipeline issues; active participant in Agile/Scrum ceremonies.

Client: Barnes & Noble | Domain: Supply Chain

- Built a reporting tool from scratch, enabling business users to generate tailored reports directly from a legacy Order Management System, with strict accuracy and uptime requirements.
- Designed and developed a delta ingestion pipeline using AWS Glue, S3, and PostgreSQL to ingest and transform data from SFTP sources, powering robust batch ETL workflows for legacy systems.
- Built serverless APIs on AWS Lambda to enable on-demand report generation and scheduling in XLSX format, significantly improving reporting efficiency and user experience.
- Engineered the organization's first custom reporting solution, resolving critical technical challenges to establish a scalable reporting framework reused and maintained across future initiatives; documented pipeline design for team-wide reuse.
- Improved supply chain data orchestration and automation using AWS-native services, reducing manual intervention.

Client: Surescripts LLC | Domain: Healthcare IT

- Developed data solutions for doctors, pharmacies, and insurance providers using Scala and GCP-based applications to streamline healthcare data workflows, applying strong Scala programming fundamentals to production-grade pipelines.
- Built integrations with EMR and EHR systems, ensuring compliance with healthcare data standards, data quality, and delivering high-quality, actionable insights to analytics stakeholders.
- Built secure authentication systems and APIs to power personalized recommendations within the healthcare ecosystem; guided junior engineers on Scala coding practices and code reviews.

Gen AI Labs — Internal Generative AI Initiatives

- Image-Based Rack Awareness & Restocking: Developed a computer vision pipeline using PyTorch, YOLOv5, and Roboflow to analyze retail rack imagery, detect inventory levels, and trigger automated restocking alerts, reducing manual inventory checks.
- Custom Document-Query Chatbots: Engineered both commercial and open-source RAG chatbot solutions using OpenAI, LangChain, Streamlit, and Hugging Face models, with FAISS for vector storage; integrated Guardrails and the Ollama framework to improve response accuracy, security, and reliability for enterprise use cases.

Member Technical Staff – Python | OpsVeda Asia October 2020 – March 2022

- Developed and optimized supply chain data solutions for a SaaS platform, including real-time monitoring systems, automated Python-based reporting workflows, and robust data pipelines using PostgreSQL, Liquibase, and Jenkins.
- Contributed to core product engineering with bi-weekly feature releases and bug fixes, ensuring continuous improvement.

Data Engineer (Intern) | Reliance Industries June 2019 – May 2020

- Improved ETL performance by 75% through optimization of Sqoop workflows and Scala code.
- Improved query response times by 40% by tuning Hive and Phoenix configurations.

Software Engineer Trainee | Value Global April 2017 – May 2018

- Built SQL & PL/SQL data solutions for clients in the Oil & Gas domain.

EDUCATION

M.E. in Big Data & Data Analytics — Manipal School of Information Sciences, Manipal University

June 2018 – July 2020 | CGPA: 9.05/10

B.E. in Electronics and Communication Engineering — Sri Venkateswara College of Engineering (SVCE), Anna University

August 2012 – May 2016

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02) — Oct 2024
- Databricks Lakehouse Fundamentals — Feb 2024
- LangChain & Vector Databases in Production — Oct 2023
- Dataiku ETL Advanced Designer — Jan 2023
- Oracle Certified Professional (PL/SQL Developer) — Jan 2017